

|   |     |                                                                                                                             |                                                                                                                                                                                                                                                                                                                           |  |
|---|-----|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| 5 | (a) | The electric current $I$ (conventional current) at a particular point is the <u>rate of flow of positive charges</u> . [B1] |                                                                                                                                                                                                                                                                                                                           |  |
|   | (b) |                                                                                                                             |                                                                                                                                                                                                                                                                                                                           |  |
|   |     | (i)                                                                                                                         | Total amount of charge due to the free electrons in the wire = $ALne$ [B1]                                                                                                                                                                                                                                                |  |
|   |     | (ii)                                                                                                                        | <p>Time taken for the total amount of charge to move through the wire = <math>L/v</math> [B1]</p> <p>Current = rate of flow of charge = <math>\frac{ALne}{\frac{L}{v}}</math> [B1]</p> <p>= <math>nAve</math> [A0]</p> <p>Alternative use <math>I = Q/t</math> [B1], explain <math>v = L / t</math> [B1]</p>              |  |
|   | (c) |                                                                                                                             |                                                                                                                                                                                                                                                                                                                           |  |
|   |     | (i)                                                                                                                         | <p>Since current is the same throughout the wire</p> <p><math>nA_X v_X e = nA_Y v_Y e</math> [C1]</p> <p><math>\frac{v_Y}{v_X} = \frac{A_X}{A_Y} = \frac{\pi(\frac{d_X}{2})^2}{\pi(\frac{d_Y}{2})^2} = (\frac{d_X}{d_Y})^2 = (\frac{1}{0.69})^2 = 2.1</math> [A1] (student shows ratio of square of diameter suffice)</p> |  |
|   |     | (ii)                                                                                                                        | <p>As the diameter decreases, the resistance across the damaged part of the wire increases. [M1]</p> <p>As such, the power dissipated in the damaged length of wire increases. [A1]</p>                                                                                                                                   |  |

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|---|-----|----------------------------------------------------------|--|----|
| 6 | (a) | magnitude of induced electromotive force proportional to |  | B1 |
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